

LLM 2-dars

Token vs Words:

1 token $\approx$ 0.75 words. (default)

- A chunk of text (characters, sub-words or symbols) processed by LLM
- Numerical vector ID stored in the model's vocab matrix
- Can be a word, sub-word or punctuation

Tokenlarda 3ta case Bolishi mumkin:

• Token = Words : Cats sit on a mat

• Token > Words : Decons di/rutinalization

• Token < Words : United States $\rightarrow$ [UnitedStates]

Types of Tokens:

- **Whole Word Tokens:** 1 word 1 Token, cat = [cat]
- **Sub word Tokens:** Murakkab sozlarni bo'laklarga ajratgan holda tokenlash.
unbelievable. [un] + [able].
- **Character & Fragment Tokens:** Typos, rare codes, unique strings are broken down all the way to single characters.
World = [w] + [o] + [rld].
- **Non-Word Elements:** Punctuations, spaces, formatting

Why LLMs ~~are used~~? use token

- Vocabulary control:

- Immunization to typos: misspelled word

- Compute & Cost tracking: represent GPU compute

Tensorlar: $N \times N$ u-n maxsus data structure

Hisobga olish kerak:

- Parameters: $12 * L * d^2$
- Xotira: $16 B * N$
- Compute byudjet: $6 * N * D$
- Vaqt: $6 * N * D / GPU * peak * MFU$
- Narx: soat * GPU. \$
- Samoradorlik: $MFU * BF16 * Checkpoint$

Arxitektura: Parametrlarni xisoblash

Layer (L): transformer layerlar soni

d_{model} (d): Model kengligi

kontest uzunligi

Optimizer

Precision

Hardware

N

Sigim

16 bayt/parameter

- Weights (FP 32) 4B
- Gradients (FP32) 4B
- Adam m $\rightarrow$ yönəliş 4B
- Adam v $\rightarrow$ kəttəlik 4B

Parameter

N

$$16B \times 1.21B \approx 19.3GB$$

Compute

$$D \approx 20 * N$$

$$C \approx G * N * D$$

6ND

Vəqf

$$t = \frac{G * N * D}{GPU * Peak * MFU}$$

$$GPU * Peak * MFU$$

Cost

$$Soat * GPU * \$$$

$$Misal \quad 97 soat * 4 GPU * \$1.5 - 2 / soat$$

Samoradkorlik

- MFU ni oshirish
- BF16
- Activation checkpoint
- Batch / shape kurlash

Trade off → biror narsaga erishish uchun nimasidan
dir erish

FLOP

- FB 32 → 32 bit → GOLD Standard
- FP 16 → 16 bit → Gradient hishlagandagi xatolik
- BF16 → 16 bit ≈ FB32

Tensor operations:

Stride: Tensorlarning har bir o'lchovi
Böylab bir qadam oldinga siljish uchun noldan
memory/pozitsiya kerakligini ko'rsatadigan tuple
Bir storageni turli shape lar bilan baham
korishni imkonini beradi

Tensor joylashuvi tezlikni belgilaydi.

Contiguous tensor → Elementlarni memoryda
1ta uzluksiz blokda row-major (C-style) tartibida
saqlanadigan tensor

Non Contiguous → $x.T$

Einops → Matrix multiply, dot product, outer product
va tensor contraction/ni string uslubiga,
bu birlashtirilgan umumlashma

- Self documenting - o'lchov nomlari niyatni
izohsiz aniq qiladi.

MFU → model FLOPs Utilization. ↓

Sekundiga bajarilgan foydali model FLOPs
ning nazariy peak hardware FLOPs ga nisbati
GPU qay darajada samarali qaydalaniboli

nn.Parameter → nn.Parameterga o'ralgan tensor avtomatik ravishda module ning o'rganiladigan qismiga parametri sifatida royxatdan o'tadi, — model.parameters() da paydo bo'ladi, gradient hisobiga kiradi.

Xavier Init (2010) → Weightlarni $1/\sqrt{d-init}$ da shakllantiradi. Vanishing / Exploding oldini oladi

Training loop: zero_grad() → forward → loss → backward → clip_grad_norm(2.0) → optimizer.step() → checkpoint

UNK → Unknown token, used in a word tokenization when word is not in vocab